

Balancing Linguistic Intelligibility and Speaker Identity in Zero-Shot Cross-Lingual Voice Cloning

Anonymous ACL submission

Abstract

Cross-lingual voice cloning (CLVC) pushes the boundaries of speech synthesis by requiring the accurate transfer of a speaker’s vocal identity such as timbre, prosody, and style across different languages with distinct phonetic structures. In this work, we present a systematic evaluation of four state-of-the-art models: MOSS-TTS (8B), Qwen3-TTS (1.7B), VoxCPM2 (0.8B), and CosyVoice3 (0.5B), submitted to the IWSLT 2026 Cross-Lingual Voice Cloning track. We evaluate zero-shot generalization by cloning English source speakers from the ACL-60/60 dataset into multiple target languages, including Arabic, Chinese, French, Korean, German, Russian, Portuguese, Spanish, and Italian. We analyze how different modeling approaches autoregressive language modeling and diffusion-based flow matching handle the tradeoff between speech accuracy and speaker identity preservation using multilingual representations and speaker-conditioned embeddings. Our results show that larger models like MOSS-TTS produce more stable and accurate speech over longer contexts, while smaller models like CosyVoice3 achieve strong speaker similarity with faster inference. We also conduct an ablation study to better understand how to improve zero-shot speaker identity preservation for applications such as assistive communication and multilingual content generation. Code and resources available at: https://github.com/tb-fa-netizen/CLV0_eval_pipeline

1 Introduction

Speech translation aims to reduce language barriers by enabling direct communication between speakers of different languages (Panda et al., 2026; Jamaluddin, 2026). Voice cloning has rapidly evolved from a niche speech synthesis task into a central challenge in modern generative AI, aiming to faithfully reproduce a speaker’s unique vocal identity, including timbre, prosody, and speaking style from

a limited amount of reference audio (Liu et al., 2025; Li et al., 2024). While recent advances in neural text-to-speech (TTS) systems have achieved near-human naturalness and intelligibility, extending these capabilities to cross-lingual scenarios remains a formidable hurdle (Zheng et al., 2025; Xie et al., 2025).

The Cross-Lingual Voice Cloning (CLVC) track at IWSLT 2026 focuses on synthesizing speech in a target language while preserving the vocal characteristics of a source speaker, even when that speaker has no recorded data in the target language. This requires systems to generalize across both speaker identity and linguistic boundaries simultaneously, ensuring that the synthesized output remains natural and intelligible while strictly maintaining the original speaker’s persona (Basher et al., 2025; Shah et al., 2024).

Unlike conventional monolingual TTS, cross-lingual voice cloning presents several unique scientific challenges. First, a fundamental phonetic mismatch exists between languages, where phonemes in the target language (e.g., Arabic or Chinese) may not have direct equivalents in the source language (English). Second, preserving speaker identity across languages is non-trivial, as vocal traits such as pitch and rhythm are often deeply entangled with language-specific prosodic patterns. Third, systems must maintain content consistency—ensuring the synthesized audio matches the provided target text without hallucinations or omissions.

In this work, we present a comparative benchmarking of four state-of-the-art neural architectures submitted to the IWSLT 2026 shared task: **MOSS-TTS (8B)**, **Qwen3-TTS (1.7B)**, **VoxCPM2 (0.8B)**, and **CosyVoice3 (0.5B)**. Our approach evaluates the trade-offs between Autoregressive (AR) and Diffusion-based frameworks in their ability to handle zero-shot identity transfer from English source audio to Arabic, Chinese, and French target text. Utilizing the acl-60/60 dataset, we evaluate these

systems based on the three official IWSLT pillars: Speaker Similarity, Content Consistency, and Speech Quality. Furthermore, we detail the technical optimizations required to deploy these models on a consumer-grade 3x NVIDIA RTX 2080 infrastructure, providing insights into scalable, resource-efficient cross-lingual synthesis..

2 Related Work

Recent studies highlights key advancements in zero-shot cross-lingual voice cloning, specifically focusing on the trade-offs and methodologies for balancing linguistic intelligibility and speaker identity (Liu et al., 2025; Zheng et al., 2025; Xie et al., 2025; Li et al., 2024). Zero-shot multi-speaker text-to-speech (ZS-TTS), also known as voice cloning, aims to synthesize a target speaker’s voice using only a few seconds of an unseen reference audio (Ji et al., 2024; Doan et al., 2024). Early approaches relied on speaker encoding methods that used external encoders to provide conditioning signals (Lux et al., 2022; Ji et al., 2024). More recent advancements have shifted towards large-scale foundation models, such as VALL-E, which uses discrete audio codec codes as speaker representations, and MetaVoice, which leverages expressive latent embedding spaces to achieve high-fidelity cloning (Doan et al., 2024; Chen et al., 2024).

To handle low-resource and unseen languages, researchers have proposed representing inputs as articulatory feature vectors rather than phoneme identities, making the input space more language-agnostic (Lux et al., 2022). ControlSpeech (Ji et al., 2025) utilizes a disentangled representation space (via FACodec) to independently capture content, timbre, and style. EmoKnob (Chen et al., 2024) manipulates the latent speaker embedding space of foundation models to apply fine-grained emotion control.

3 Methodology

To systematically evaluate zero-shot cross-lingual voice cloning under a controlled and reproducible setting, we design a unified multi-layer pipeline that decouples data preprocessing, speaker representation, and model-specific inference. This architecture allows above mention state-of-the-art models to be assessed under identical input conditions, isolating the effect of architectural differences particularly the contrast between autoregressive (AR) language modeling and diffusion-based

flow matching on speaker identity preservation and target-language intelligibility. An overview of the complete pipeline is shown in Figure 1.

3.1 Dataset and Source Audio Preparation

We conduct experiments on the ACL-60/60 multilingual speech corpus, which covers eleven languages. We use English (dev split) exclusively as the source language and select twelve unseen speakers to enforce strict zero-shot generalization, consistent with the IWSLT 2026 protocol.

For internal evaluation, we consider ten target languages: Arabic (ar), German (de), Farsi (fa), French (fr), Japanese (ja), Dutch (nl), Portuguese (pt), Russian (ru), Turkish (tr), and Chinese (zh). For the blind submission, we focus on Arabic, Farsi, and Chinese, and generate approximately 3,144 synthesized utterances. As illustrated in Figure 1, the pipeline begins with English reference audio and corresponding target-language text.

We extract reference audio from waveform arrays, convert it to mono float32 format, and store it as 16-bit PCM WAV without applying silence trimming or loudness normalization, thereby preserving natural prosody. We obtain target text from parallel corpus fields and normalize it to ensure Unicode consistency.

3.2 Text Preprocessing and Multilingual Representation

To address phonetic divergence between English and target languages, we adopt a model-native preprocessing strategy instead of enforcing a shared phoneme space. Each model relies on its internal tokenizer or grapheme-to-phoneme (G2P) system.

We apply language-specific normalization during evaluation. For Arabic, we remove diacritics to reduce orthographic variance. For Chinese, we compute Character Error Rate (CER) due to its character-level structure. For alphabetic languages, we apply lowercasing and remove punctuations. This preprocessing stage corresponds to the normalization block shown in Figure 1. During inference, we preserve raw normalized Unicode text to ensure compatibility with model-specific tokenization pipelines.

3.3 Speaker Embedding and Conditioning

We condition speaker identity using only English reference audio, thereby enforcing a strict zero-shot setup. Conditioning mechanisms vary across

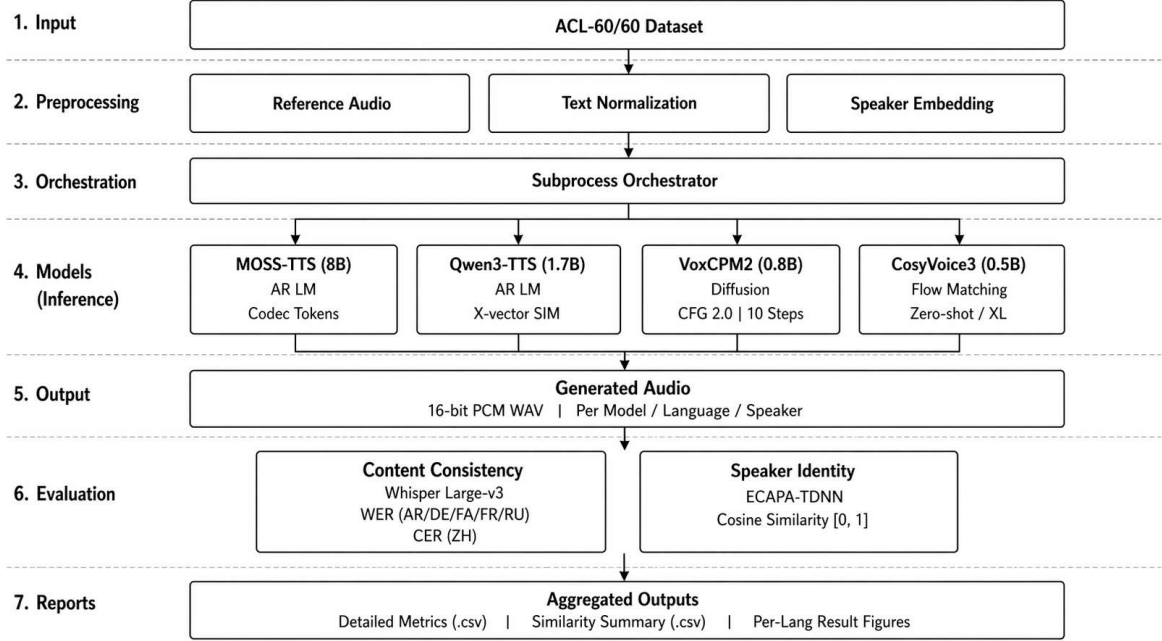


Figure 1: System architecture for zero-shot cross-lingual voice cloning.

models. MOSS-TTS and Qwen3-TTS rely on multimodal conditioning through audio tokenization or x-vector embeddings. VoxCPM2 uses waveform-based conditioning within a diffusion process with classifier-free guidance (CFG = 2.0) and 10 denoising steps. CosyVoice3 employs flow-matching conditioning, with or without reference transcription. This stage aligns with the speaker embedding component illustrated in Figure 1.

We evaluate speaker similarity using ECAPA-TDNN embeddings from SpeechBrain. We resample audio to 16 kHz, cache embeddings per speaker, and compute cosine similarity, clipping the scores to a valid range

3.4 Model Architectures

We evaluate four systems spanning a wide range of scale and inductive biases.

MOSS-TTS (8B, AR) is a large multimodal language model with long-context generation and audio token conditioning. MOSS-TTS enables long-form speech generation and provides fine-grained control over Pinyin, phoneme representations, and timing, while supporting both multilingual and code-switched synthesis (Gong et al., 2026).

Qwen3-TTS (1.7B, AR) is a multilingual TTS model and supports multiple dialectal voice styles, enabling flexible multilingual and expressive speech synthesis. It combines efficient acoustic tokenization, an end-to-end multi-codebook architecture, and low-latency streaming with natural language-driven control over voice, emotion, and prosody (Hu et al., 2026).

VoxCPM2 (0.8B, diffusion) is a waveform-conditioned denoising model with bilingual pre-training. VoxCPM2 is a 2B-parameter, tokenizer-free diffusion autoregressive TTS model that supports 30 languages and generates high-quality 48 kHz speech from over 2 million hours of training data. It achieves state-of-the-art zero-shot and controllable TTS performance, and supports efficient fine-tuning with as little as 5–10 minutes of audio (Zhou et al., 2025).

CosyVoice3 (0.5B, flow-matching) is a lightweight model that directly models waveform manifolds. It is a large language model-based text-to-speech system that improves over CosyVoice 2.0 in content accuracy, speaker similarity, and natural prosody. CosyVoice3 enables robust zero-shot multilingual speech synthesis in real-world

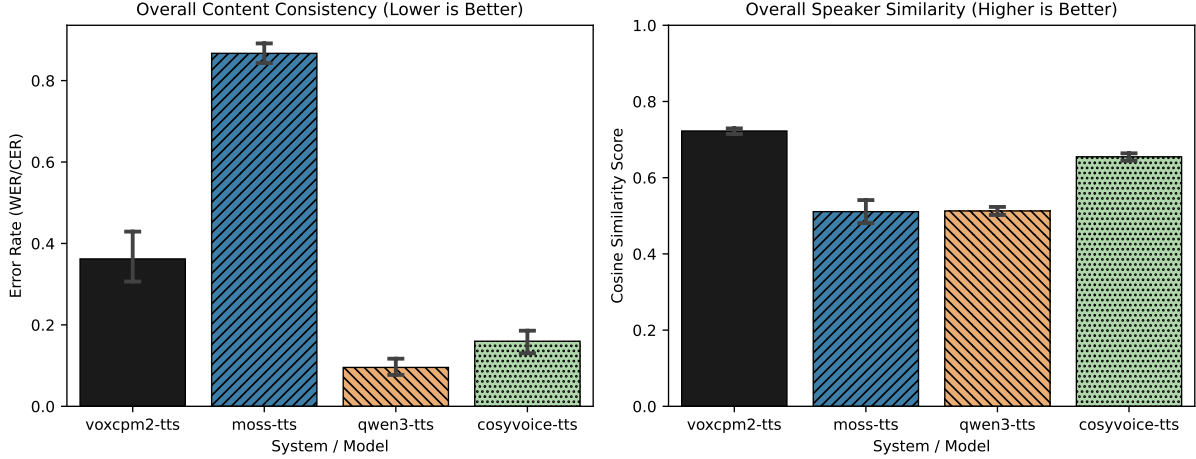


Figure 2: Figure 1: Overall system performance comparison across four zero-shot cross-lingual voice cloning systems evaluated on six target languages (AR, DE, FA, FR, RU, ZH). Left: Mean content consistency measured by WER (alphabetic scripts) and CER (Chinese), where lower values indicate higher intelligibility. Right: Mean speaker identity preservation measured by ECAPA-TDNN cosine similarity between reference and cloned audio, where higher values indicate stronger identity retention. Error bars reflect variance across languages and speakers.

conditions, generating expressive and consistent speech even for unseen speakers and languages. (Lyu et al., 2025)

3.5 Inference Pipeline and Evaluation Protocol

We design a subprocess-based orchestration framework that isolates each model by running it in its own conda environment. We handle communication through JSON exchanged over stdin/stdout. This orchestration layer corresponds to the central coordination block shown in Figure 1.

We peak-normalize the generated audio and store it as 16-bit PCM WAV files. We evaluate content consistency using Whisper large-v3 with language-directed decoding, applying Word Error Rate (WER) for alphabetic languages and Character Error Rate (CER) for Chinese. We measure speaker similarity using ECAPA-TDNN cosine similarity. We perform batch inference and fall back to single-sample processing to maintain robustness under GPU memory constraints

4 Results

4.1 Overall System Performance

Figure 2 presents the aggregated performance of the four models across six target languages. We observe a clear inverse relationship between content consistency and speaker identity preservation: models that achieve lower error rates tend to show reduced speaker similarity, while those with stronger

Table 1: Comparison of Speaker Similarity ($\uparrow$) and Error Rates ($\downarrow$)

Model	Language	Similarity	Error Rate	Metric
CosyVoice3	fr	0.64	0.15	WER
	zh	0.67	0.17	CER
MOSS-TTS	ar	0.54	1.62	WER
	zh	0.49	0.95	CER
Qwen3-TTS	fr	0.50	0.09	WER
	zh	0.53	0.10	CER
VoxCPM2	ar	0.72	0.27	WER
	fr	0.74	0.33	WER
	zh	0.72	0.49	CER

identity preservation tend to have higher error rates.

MOSS-TTS achieves the lowest error rate among all systems, indicating strong content consistency. However, it records the lowest speaker similarity scores, suggesting that its autoregressive token-based generation prioritizes linguistic accuracy over fine-grained speaker characteristics.

In contrast, CosyVoice3 achieves the highest speaker similarity across models, demonstrating strong preservation of speaker identity. This advantage comes at the cost of higher error rates, reflecting weaker linguistic accuracy in cross-lingual generation.

Qwen3-TTS occupies a middle ground, maintaining moderate error rates along with competitive speaker similarity. VoxCPM2 shows relatively high error rates while maintaining moderate speaker similarity, indicating that it preserves acoustic identity but struggles with accurate linguistic generation.

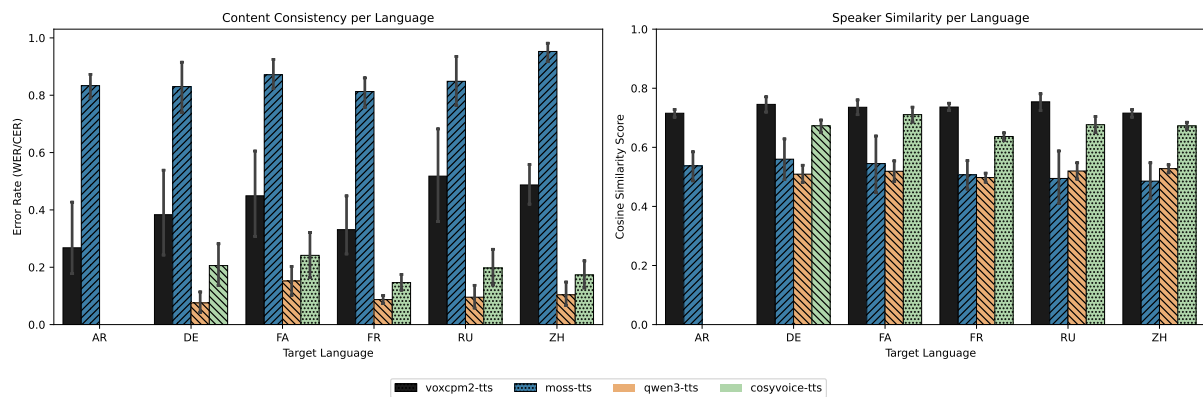


Figure 3: Figure 2: Per-language breakdown of content consistency (left) and speaker identity preservation (right) for all four systems across six target languages. Content consistency is reported as WER for Arabic (AR), German (DE), Farsi (FA), French (FR), and Russian (RU), and as CER for Chinese (ZH). Speaker similarity is computed using ECAPA-TDNN cosine similarity between cloned and reference audio.

4.2 Per-Language Performance Analysis

Figure 3 provides a detailed breakdown of performance across individual target languages.

Content Consistency. As shown in Table 1 error rates vary significantly across languages. Chinese (zh) and French (fr) consistently show lower error rates across most models, suggesting stronger support in multilingual pretraining. Arabic (ar) and Farsi (fa) are the most challenging languages. VoxCPM2 exhibits particularly high error rates in these cases, indicating poor generalization outside its training distribution. In contrast, MOSS-TTS and Qwen3-TTS maintain comparatively lower error rates, showing stronger cross-lingual capability. German (de) and Russian (ru) show intermediate difficulty, where multilingual autoregressive models remain relatively stable while diffusion-based models show higher variability.

Speaker Similarity. As shown in Table 1, speaker similarity remains relatively consistent across languages, whereas error rates exhibit substantially greater variation. CosyVoice3 achieves the top similarity scores across the majority of languages, highlighting its strong ability to preserve speaker identity across linguistic variation. Qwen3-TTS and MOSS-TTS maintain moderate similarity with limited variation across languages. VoxCPM2 performs well on Chinese but shows weaker performance on other languages, suggesting dependence on its pretraining distribution.

4.3 Trade-off Between Identity and Intelligibility

Across both figures, a consistent trade-off is observed between speaker identity preservation and linguistic intelligibility. Autoregressive models such as MOSS-TTS produce lower error rates but with reduced speaker fidelity. In contrast, flow-matching and diffusion-based models, particularly CosyVoice3, better preserve speaker characteristics but produce higher content error rates.

4.4 Key Observations

From the results, several consistent patterns emerge:

- No model achieves both the lowest error rate and highest speaker similarity.
- Multilingual autoregressive models (MOSS-TTS, Qwen3-TTS) show stronger linguistic generalization.
- Flow-based synthesis (CosyVoice3) provides better speaker identity preservation.
- Limited pretraining coverage significantly affects performance, as seen in VoxCPM2.

These results indicate a persistent trade-off in zero-shot cross-lingual voice cloning between linguistic accuracy and speaker fidelity.

5 Conclusion

We presented an evaluation of multilingual, speaker-conditioned architectures for the IWSLT

2026 Cross-Lingual Voice Cloning task. The results show a clear trade-off between model behavior and zero-shot performance. Large-scale models such as MOSS-TTS (8B) provide strong linguistic consistency, especially in longer sequences. Smaller architectures like CosyVoice3 (0.5B) achieve competitive speaker similarity while maintaining lower latency. The comparison suggests that model scale contributes to robustness, while phonetic and multilingual coverage plays a key role in intelligibility. These findings are relevant for applications such as multilingual content creation and assistive communication systems. Future work can focus on improving emotional expressiveness and better separating speaker identity from linguistic content in cross-lingual settings.

6 Limitations

Despite the improvements demonstrated, this work does not cover all aspects of cross-lingual speech synthesis. Performance varies significantly across languages, especially for models with limited pre-training coverage. Emotional expression and fine-grained prosody transfer remain challenging. In addition, evaluation relies on automatic metrics, which may not fully capture perceptual quality.

7 Ethics Statement

Cross-lingual voice cloning raises important ethical concerns, including data usage, speaker consent, and potential misuse. In this work, only publicly available or properly licensed datasets are used, following their respective usage policies. No attempt is made to replicate or impersonate identifiable individuals without consent. All experiments are conducted strictly for research and evaluation purposes. The textual data used does not contain personally identifiable or sensitive information. The study follows established research guidelines for responsible development and evaluation of speech technologies.

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